Steering LVLMs via Sparse Autoencoder for Hallucination Mitigation

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Abstract

Large vision-language models (LVLMs) have achieved remarkable performance on multimodal tasks. However, they still suffer from hallucinations, generating text inconsistent with visual input, posing significant risks in real-world applications. Existing approaches to address this issue focus on incorporating external knowledge bases, alignment training, or decoding strategies, all of which require substantial computational cost and time. Recent works try to explore more efficient alternatives by adjusting LVLMs' internal representations. Although promising, these methods may cause hallucinations to be insufficiently suppressed or lead to excessive interventions that negatively affect normal semantics. In this work, we leverage sparse autoencoders (SAEs) to identify semantic directions closely associated with faithfulness or hallucination, extracting more precise and disentangled hallucination-related representations. Our analysis demonstrates that interventions along the identified faithful direction can mitigate hallucinations, while those along the hallucinatory direction can exacerbate them. Building on these insights, we propose **S**teering LVLMs via **S**AE **L**atent Directions (SSL), a plug-and-play method based on SAE-derived latent directions to mitigate hallucinations in LVLMs. Extensive experiments demonstrate that SSL significantly outperforms existing decoding approaches in mitigating hallucinations, while maintaining transferability across different model architectures with negligible additional time overhead. The code is available at https://github.com/huazhenglin2003/SSL.

1 Introduction

LVLMs (Liu et al., 2023; Dai et al., 2023b; Liu et al., 2024b) have demonstrated impressive capabilities in jointly processing visual and textual modalities, achieving strong performance on tasks such as VQA (Antol et al., 2015) and image captioning (Li et al., 2022). However, LVLMs still suffer from hallucination (Yin et al., 2011), where the generated text does not align with the visual content. This limitation poses significant challenges to their deployment in critical applications, including medical diagnosis(Gu et al., 2024) and autonomous driving(You et al., 2024), where factual consistency and reliability are essential.

To mitigate hallucination in LVLMs, researchers explore various strategies, including refining decoding algorithms (Leng et al., 2024; Huang et al., 2024; Kim et al., 2024), incorporating external

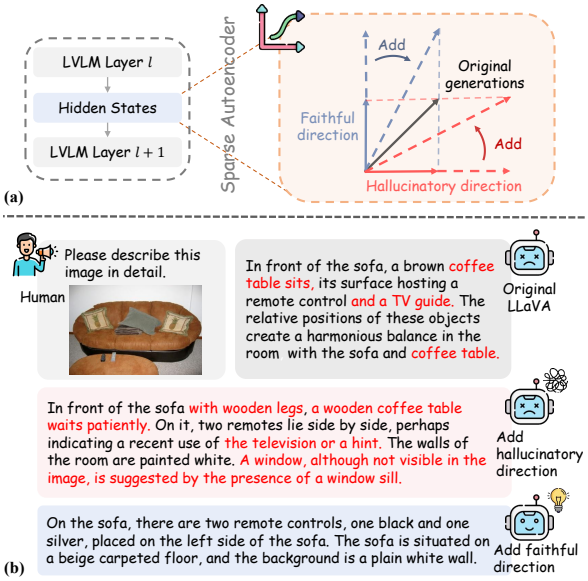


Figure 1: The figure shows, from top to bottom, the original response from the LLaVA-Next-8b, the response after intervention along the hallucinatory direction, and the response after intervention along the faithful direction. These results demonstrate that targeted interventions along faithful directions suppress hallucinatory generations, while perturbations along hallucinatory directions tend to elicit more factually incorrect content.

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Findings of the Association for Computational Linguistics: EMNLP 2025, pages 10808–10828
November 4-9, 2025 ©2025 Association for Computational Linguistics